
Fashion Compatibility Learning via Metric Learning and Modern Vision Backbones

Allizha Theiventhiram
University of Neuchâtel
allizha.theiventhiram@unine.ch

Sandra Nikoloska
University of Bern
sandra.nikoloska@unibe.ch

Boris Verdecia Echarte
University of Neuchâtel
boris.verdecia@unine.ch

Abstract

A major challenge when building an outfit is knowing how well clothing items go together. We address this as a metric learning task: given images of individual clothing items, we learn visual embeddings such that compatible items—those co-appearing in curated outfits—are nearby, while incompatible items are pushed apart. We compare two modern vision backbones (Swin-T and ConvNeXt-T) fine-tuned under triplet and contrastive losses on the Polyvore dataset. Our best model, ConvNeXt-T with triplet loss, achieves **60.8%** Fill-in-the-Blank accuracy and AUC 0.758—exceeding the type-aware supervised baseline (57%)—without any category supervision, supporting the hypothesis that modern backbone expressivity can reduce the need for explicit type conditioning.

1 Introduction and Background

Choosing items that coordinate is a non-trivial perceptual task: visual compatibility depends on color harmony, texture pairing, and stylistic coherence, with direct applications in e-commerce and virtual styling.

Problem statement. We frame compatibility as a metric learning problem. Given clothing images $\{x_1, \dots, x_n\}$, we learn $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^{128}$ such that co-outfit items are nearby while items from different outfits are pushed apart: $d(f_\theta(x_i), f_\theta(x_j)) \ll d(f_\theta(x_i), f_\theta(x_k))$, where (x_i, x_j) is a compatible pair and x_k is incompatible.

Related work. Vasileva et al. [2018] proposed type-aware embeddings with per-type projection subspaces (ECCV 2018), achieving strong results on 68,306 outfits—but requiring explicit type labels at train and test time. Tan et al. [2019] address this by learning similarity conditions as latent variables without supervision (ICCV 2019)—our most direct inspiration. Sarkar et al. [2023] model full-outfit compatibility via self-attention, but require full outfit context, less suited to single-query retrieval.

Contributions. (1) We benchmark Swin-T and ConvNeXt-T under a unified metric learning pipeline on Polyvore. (2) We compare triplet and contrastive losses under identical conditions. (3) ConvNeXt-T + Triplet achieves 60.8% FITB and AUC 0.758, surpassing the supervised baseline Vasileva et al. [2018] without type labels, to enable compatibility-aware retrieval.

2 Method

Figure 1 provides a full overview of our system.

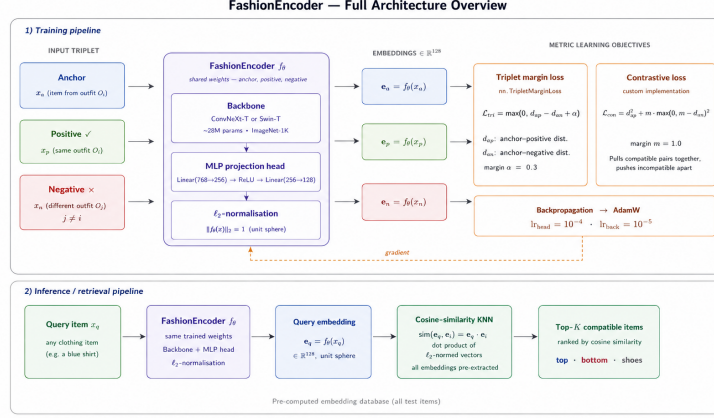


Figure 1: **FashionEncoder full architecture.** *Top (Training):* An (anchor, positive, negative) triplet passes through the shared **FashionEncoder**—a pretrained backbone followed by an MLP projection head and ℓ_2 -normalization—producing 128-dim unit-sphere embeddings. A metric loss is computed and gradients update f_θ via AdamW with differential learning rates. *Bottom (Inference):* A query item is encoded and matched against a pre-computed embedding database via cosine-similarity KNN.

2.1 Model Architecture

The **FashionEncoder** f_θ projects all item images into $\mathbb{R}^{128}$ using shared weights regardless of category. Two ImageNet-1K pretrained backbones (~ 28 M parameters each) are compared: **Swin-T** Liu et al. [2021], a hierarchical Vision Transformer with shifted-window attention; and **ConvNeXt-T** Liu et al. [2022], a modernized CNN with Transformer design choices. Tiny variants are used for GPU memory compatibility. The backbone feeds a two-layer MLP head ($768 \rightarrow 256 \rightarrow 128$, ReLU, ℓ_2 -normalization), so dot products equal cosine similarities.

2.2 Metric Learning Objectives

Training uses **(anchor, positive, negative)** triplets from outfit groupings: anchor and positive from the *same* outfit; negative from a *different* outfit.

Triplet Margin Loss (`nn.TripletMarginLoss`, $\alpha=0.3$):

$$\mathcal{L}_{\text{triplet}} = \max(0, \|f_\theta(x_a) - f_\theta(x_p)\|_2 - \|f_\theta(x_a) - f_\theta(x_n)\|_2 + \alpha) \quad (1)$$

Contrastive Loss (margin $m=1.0$):

$$\mathcal{L}_{\text{contr}} = d_{ap}^2 + \max(0, m - d_{an})^2 \quad (2)$$

2.3 Training Setup

Dataset. Marqo/polyvore (HuggingFace), 94,096 items from Polyvore Vasileva et al. [2018]. We apply an *outfit-disjoint split* (80/10/10 by outfit ID): 17,269/2,159/2,159 train/val/test outfits, preventing data leakage. The official Polyvore JSON splits are incompatible with this image subset, so literature numbers serve as order-of-magnitude reference only.

Preprocessing. Resize 224×224 ; ImageNet normalization; train augmentations: horizontal flip, color jitter.

Optimizer. AdamW, differential LRs: 10^{-4} (head), 10^{-5} (backbone); weight decay 10^{-4} ; batch 64; max 30 epochs. Two-phase training: backbone frozen for 3 warmup epochs, then full fine-tuning. Early stopping (patience 5) on val loss. Mixed precision (`torch.amp.autocast`) for $\sim 2 \times$ GPU speedup.

3 Results

Metrics. **AUC:** compatible vs. random pairs by cosine similarity. **FITB:** select correct missing item from 4 candidates given outfit context. **Recall@10:** fraction of compatible items in top-10; inherently low due to small outfit sizes (2–4 items), reported for completeness.

Table 1: Results on Polyvore test set (outfit-disjoint split, 9,430 items, 2,159 outfits). [†]Literature uses official splits; not directly comparable.

Method	AUC	FITB	Recall@10
ConvNeXt-T + Triplet (ours)	0.758	60.8%	1.25%
ConvNeXt-T + Contrastive (ours)	0.714	56.2%	0.96%
Swin-T + Triplet (ours)	0.722	54.6%	1.44%
Swin-T + Contrastive (ours)	0.710	52.4%	1.62%
Type-Aware Vasileva et al. [2018] [†]	~0.88	~57%	–
Unsup. Cond. Tan et al. [2019] [†]	~0.91	~67%	–

Discussion. All configurations substantially exceed random baselines (AUC $\gg$ 0.50, FITB $\gg$ 25%), confirming that fine-tuning on outfit co-occurrence teaches real compatibility signals. **ConvNeXt-T + Triplet wins on both AUC (0.758) and FITB (60.8%)**—a single configuration dominates across all metrics. Triplet loss consistently outperforms contrastive loss across both backbones and both metrics; no metric-dependence is observed. Backbone differences are small (ConvNeXt-T better by $\sim$ 3 pp AUC on average), indicating the training signal dominates architecture choice at this scale. Most strikingly, 60.8% FITB *without type labels* exceeds the type-aware supervised baseline (57%) by 3.8 pp, directly supporting our central hypothesis.

4 Conclusion

ConvNeXt-T + Triplet achieves 60.8% FITB and AUC 0.758 without type supervision, exceeding the type-aware supervised baseline (57%), confirming that modern backbone expressivity enables compatibility-aware retrieval without explicit type conditioning.

Limitations. Custom split prevents direct literature comparison; random negative sampling used; results have run-to-run variance (FITB/AUC sampled over 500/2000 pairs).

Future work. Semi-hard negative mining; text–image fusion with Polyvore item descriptions; evaluation on official Polyvore-Outfits splits.

References

- Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.
- Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A ConvNet for the 2020s. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- Rohan Sarkar, Navaneeth Bodla, Mariya I. Vasileva, Yen-Liang Lin, Anurag Beniwal, Alan Lu, and Gérard Medioni. OutfitTransformer: Learning outfit representations for fashion recommendation. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 3590–3598, 2023.
- Reuben Tan, Mariya I. Vasileva, Kate Saenko, and Bryan A. Plummer. Learning similarity conditions without explicit supervision. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.
- Mariya I. Vasileva, Bryan A. Plummer, Krishna Dusad, Shreya Rajpal, Ranjitha Kumar, and David Forsyth. Learning type-aware embeddings for fashion compatibility. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 390–405, 2018.